

Neural networks

Weekend 1, Friday afternoon

Carlos Cotrini

Friday 4 September 2026

1

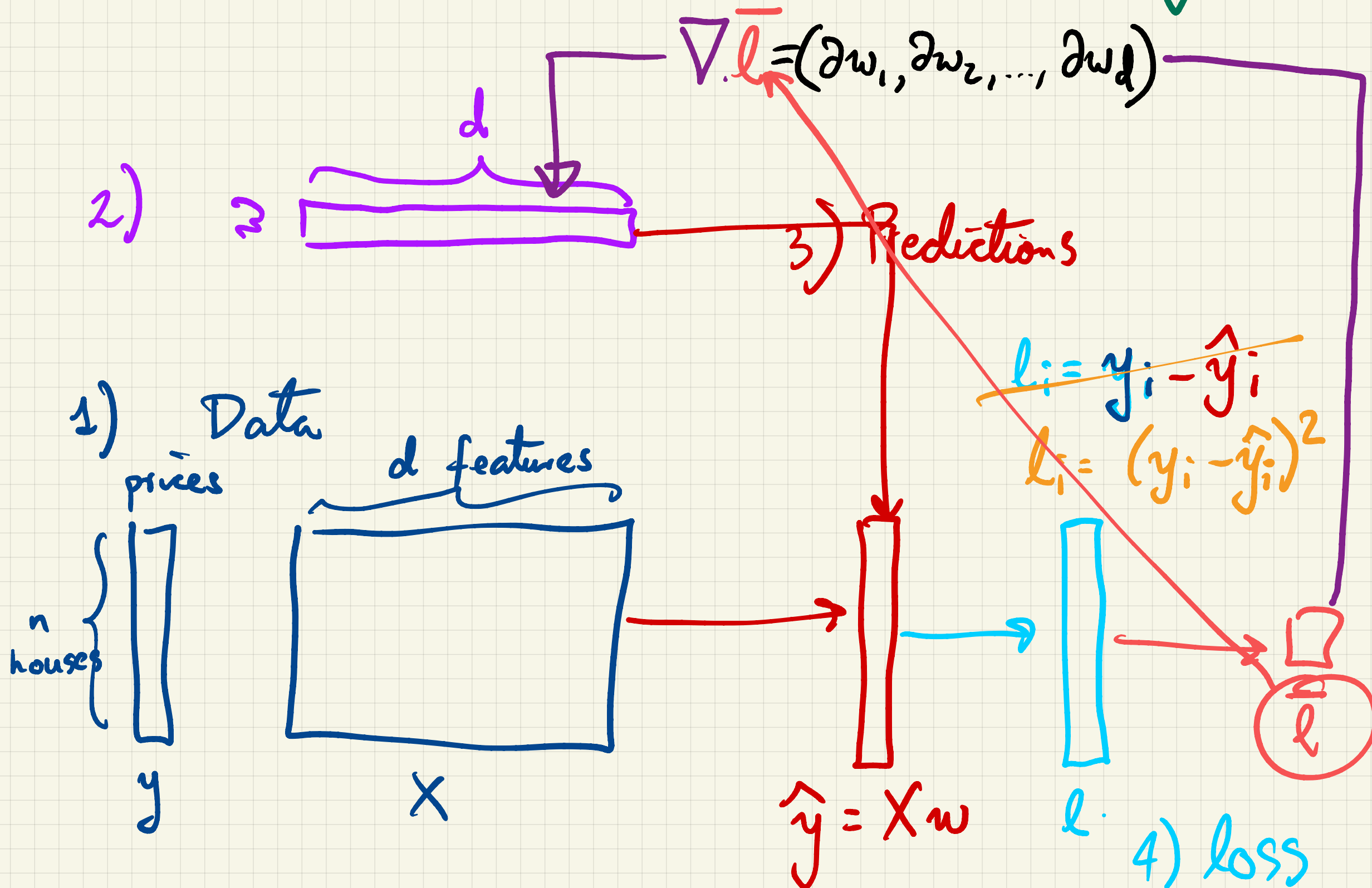
Section 1

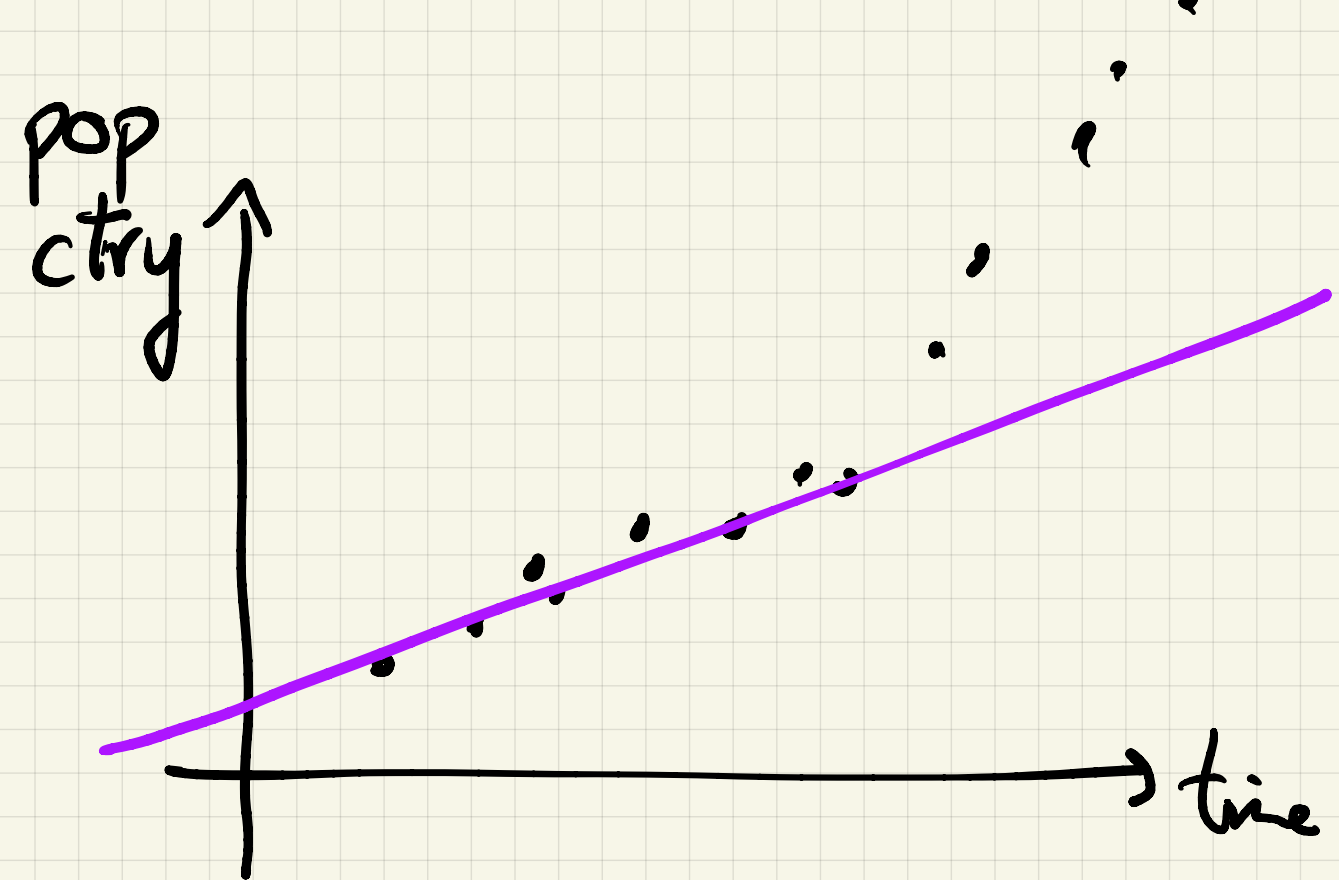
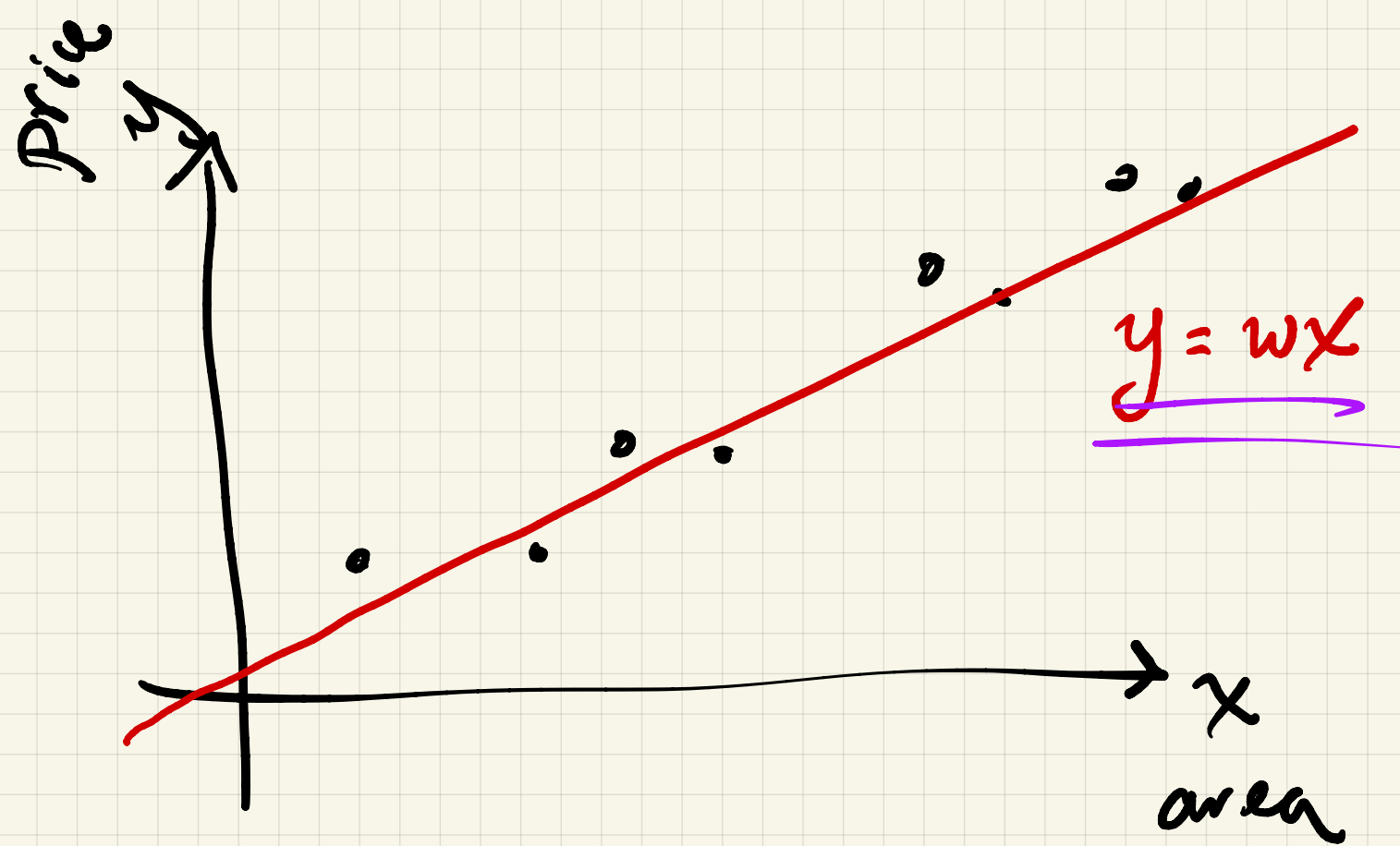
Linear regression in one picture

House prediction

w/ and

Gradient descent
lin. reg.



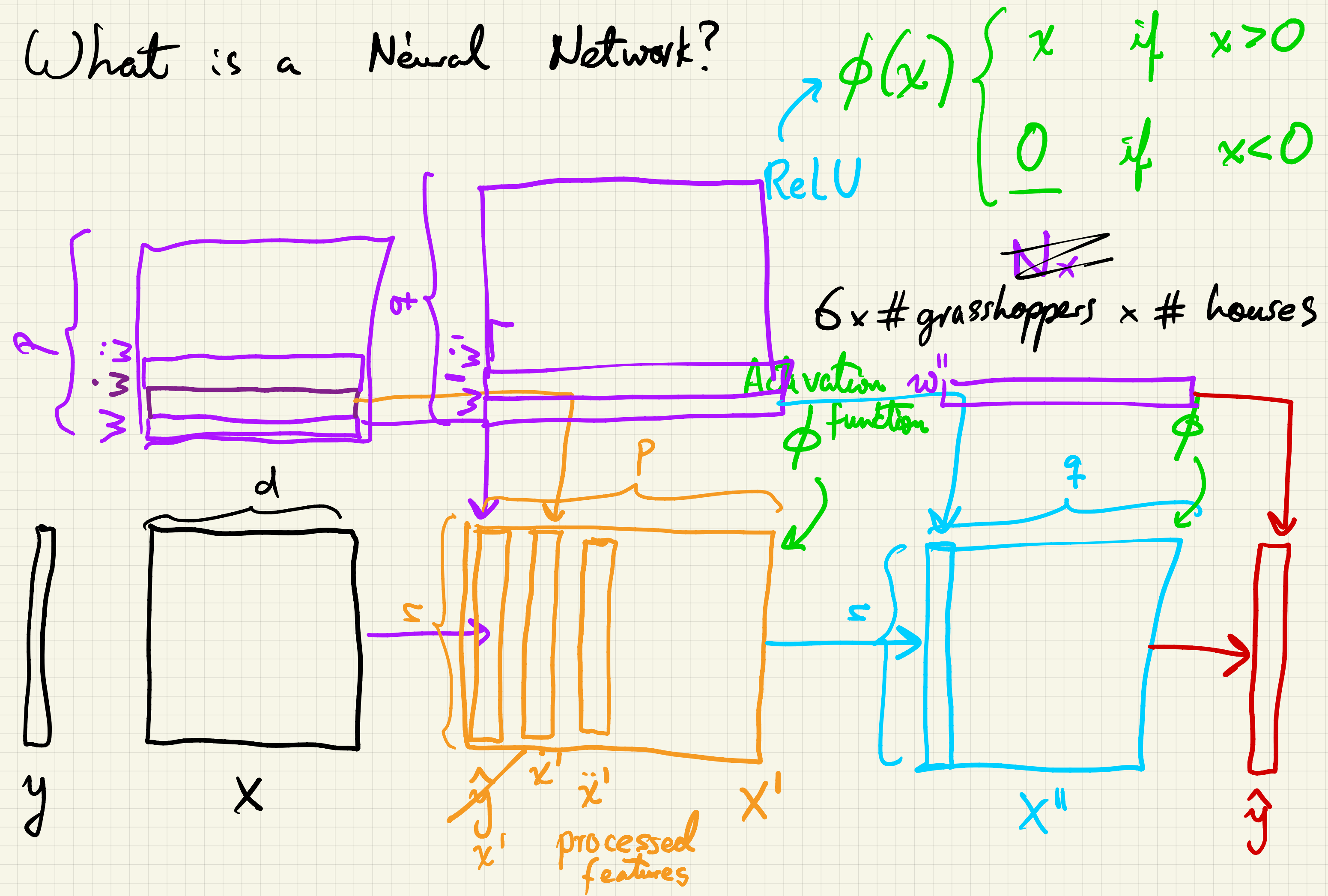


Universal
approx

- 1) Polynomials
 $y = wx + w \dots$
- 2) Neural networks

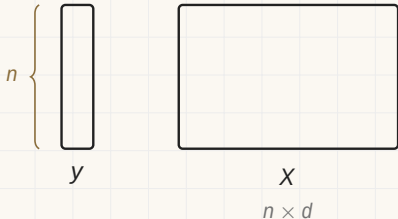


What is a Neural Network?



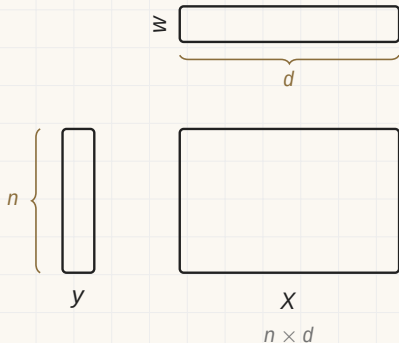
Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



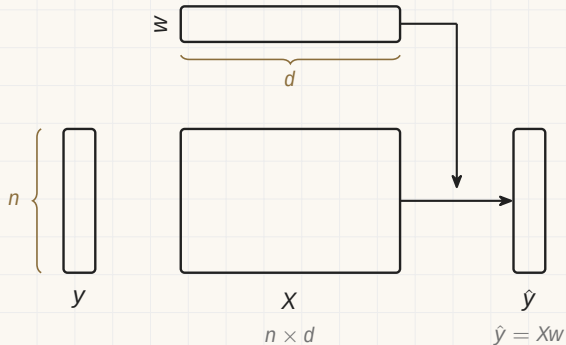
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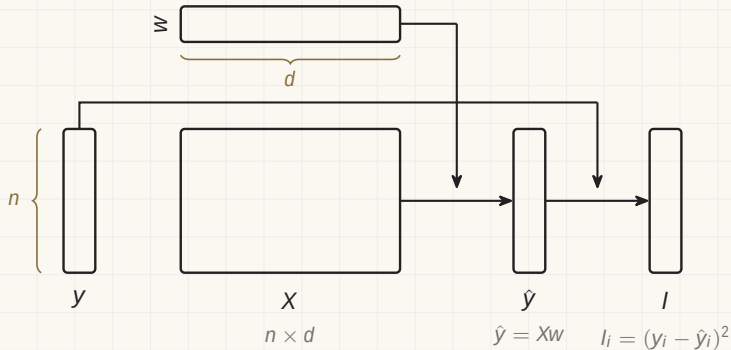
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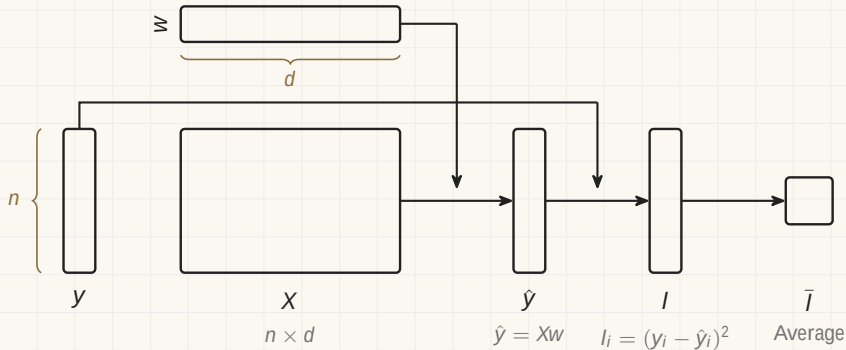
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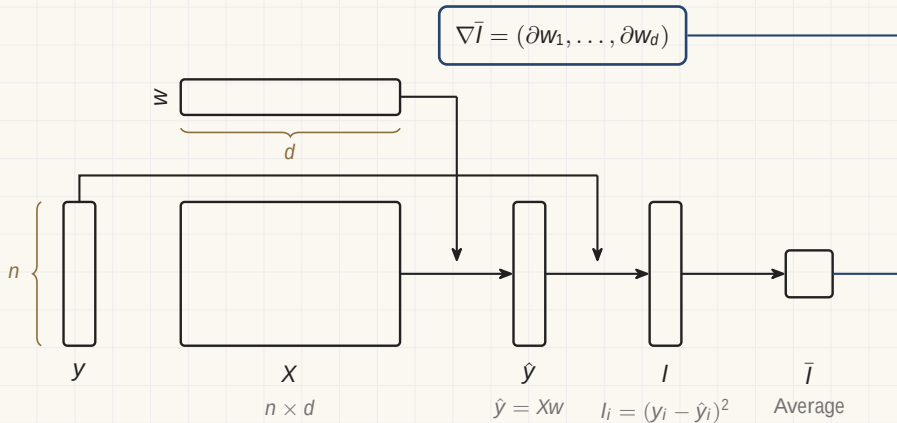
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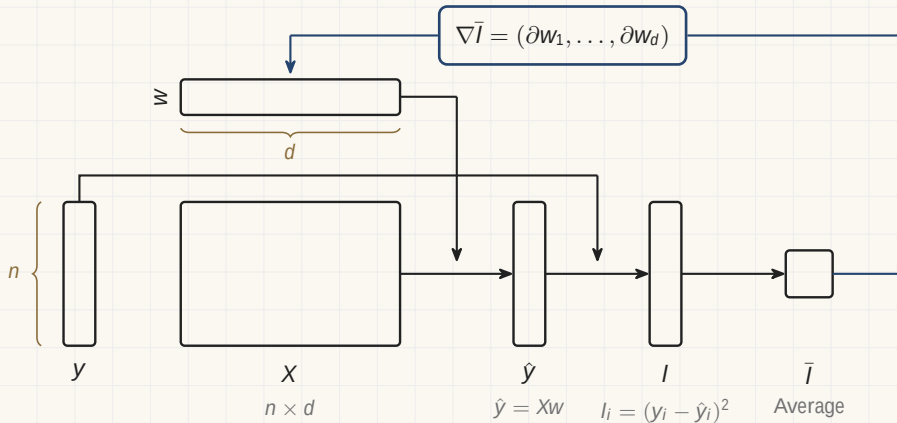
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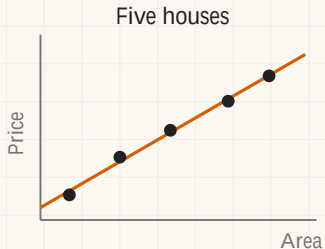
2

Section 2

What a linear model cannot do

A linear model does not $\square$ t everything

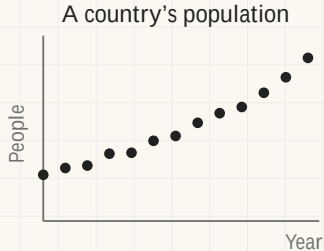
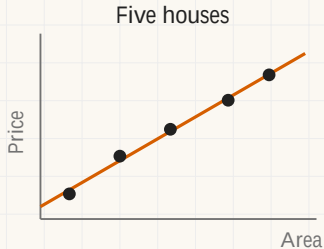
Three sets of measurements, and the best straight line through each.



The largest miss is 83'413 CHF,
which is 3.8 percent of the
price range.

A linear model does not fit everything

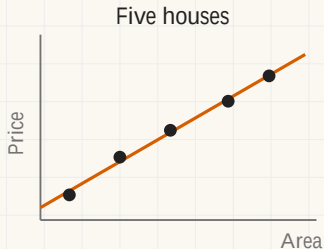
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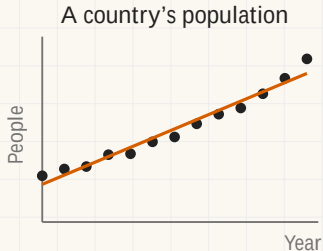
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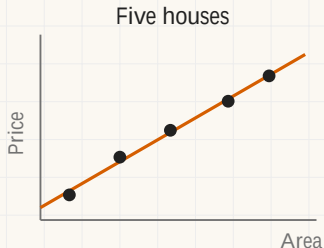
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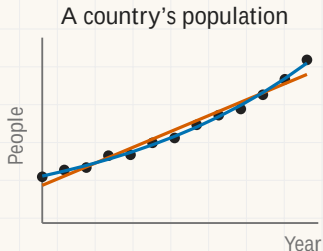
The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.

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Three sets of measurements, and the best straight line through each.



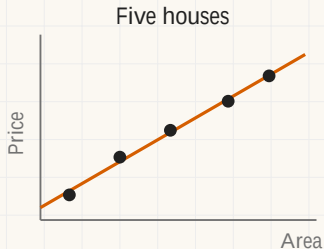
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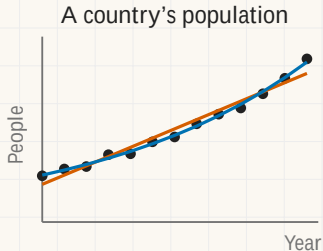
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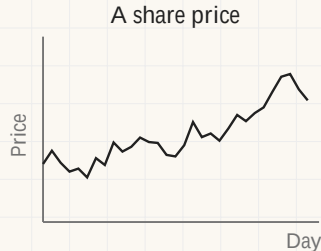
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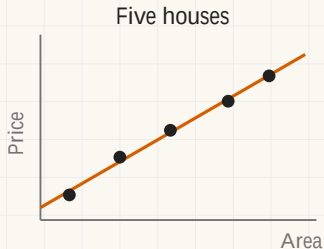


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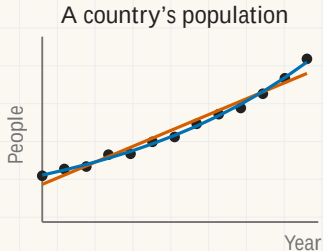


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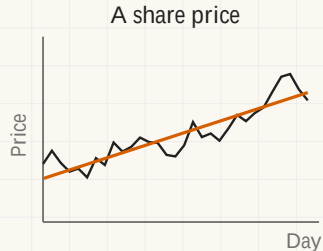
Three sets of measurements, and the best straight line through each.



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The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.



The straight line misses by 32.8 percent of the range, and no elementary curve does better. Synthetic data.

3

Section 3

Universal approximators

Universal approximators

Let K be a closed and bounded set of inputs, and let $\mathcal{F}$ be a family of functions from K to the real numbers.

$\mathcal{F}$ is a **universal approximator** on K if for every continuous $g : K \rightarrow \mathbb{R}$ and every $\varepsilon > 0$ there is an $f \in \mathcal{F}$ with $\max_{x \in K} |g(x) - f(x)| < \varepsilon$.

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- 1 The polynomials, on a closed interval. That is the Weierstrass approximation theorem, and it dates from 1885.
- 2 The neural networks. The rest of this lecture builds one, and the last slide states the theorem.

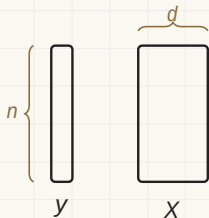
4

Section 4

Neural networks

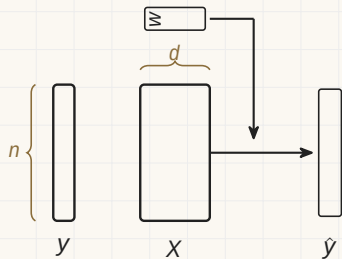
From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.



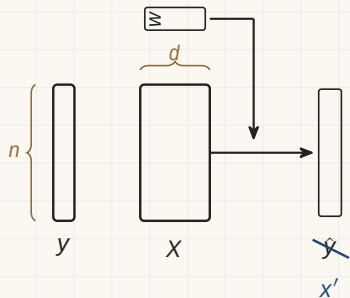
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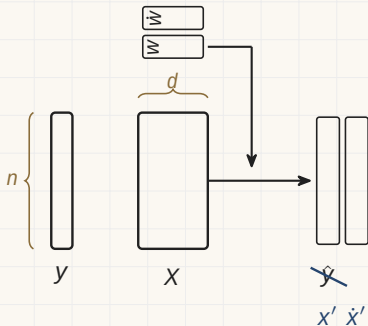
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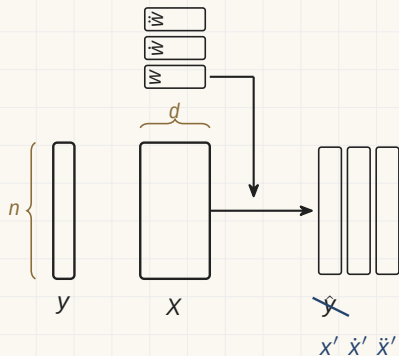
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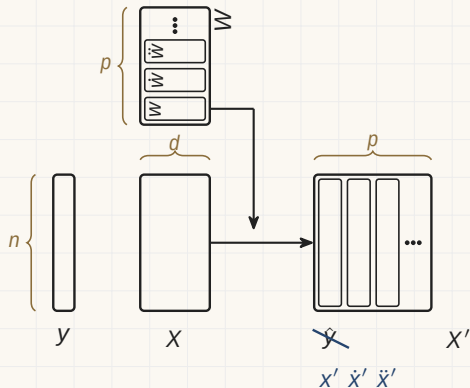
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From one weight vector to a layer

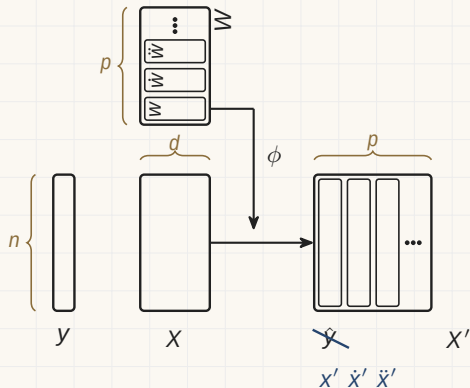
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W is laid on its side, so one of its columns is a row up there. Each row makes one column of X' , and the two are drawn at the same pitch, so you can count them off against each other. W has d rows, which is what makes the product defined, and p columns, which is a number you choose.

From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.

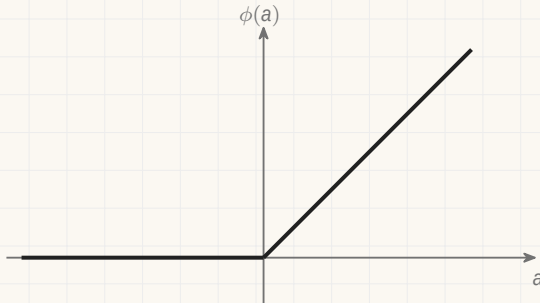


$$X' = \phi(X W)$$

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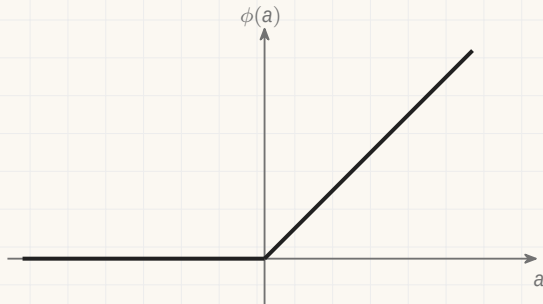
The activation function

ϕ is applied to every entry on its own, and it is the only piece of the network that is not a matrix product.



The activation function

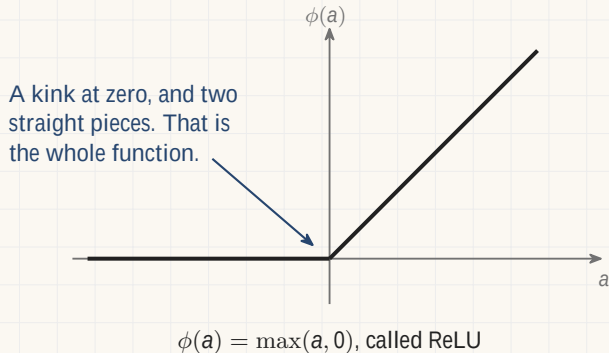
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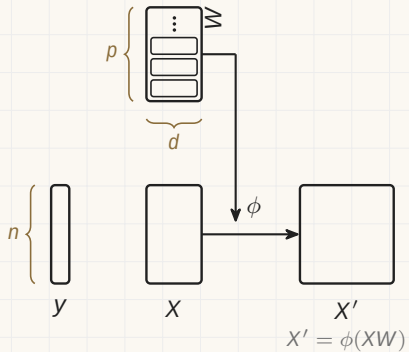
$$\phi(a) = \max(a, 0), \text{ called ReLU}$$

The activation function

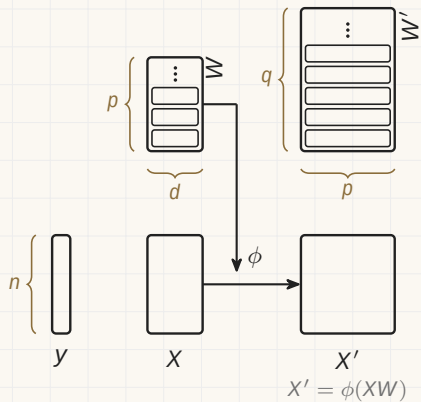
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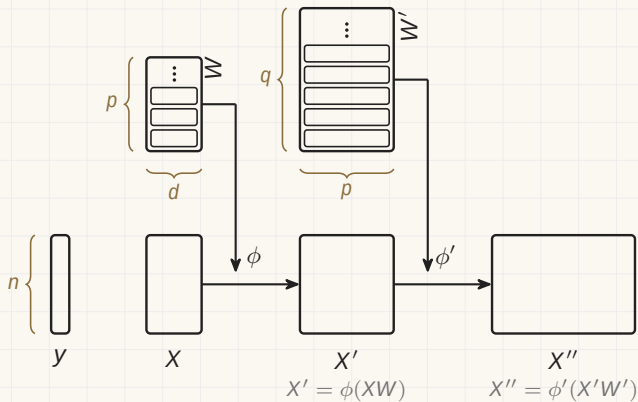
A neural network



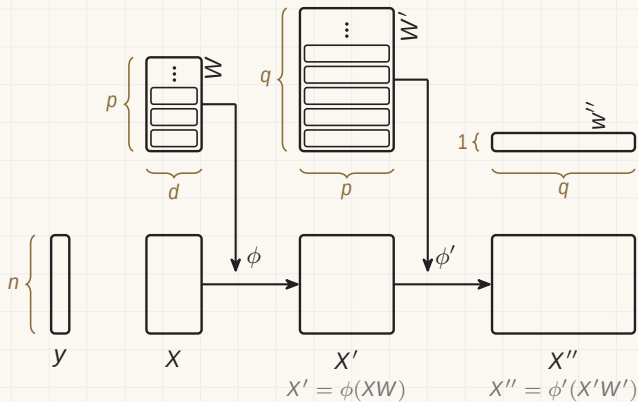
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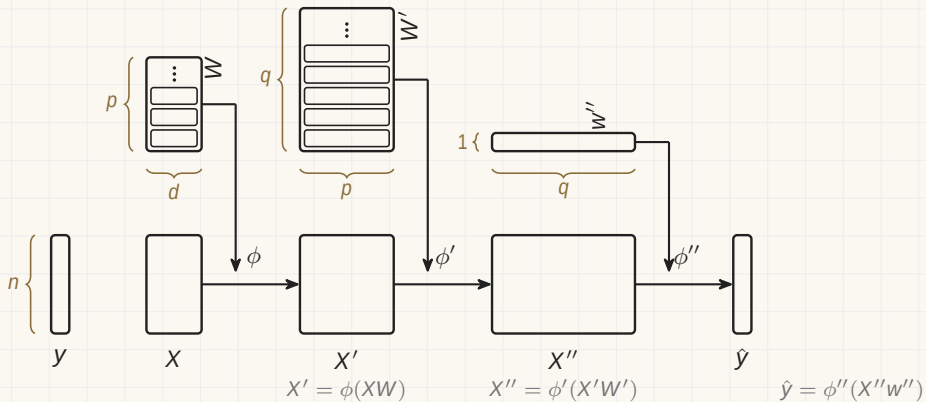
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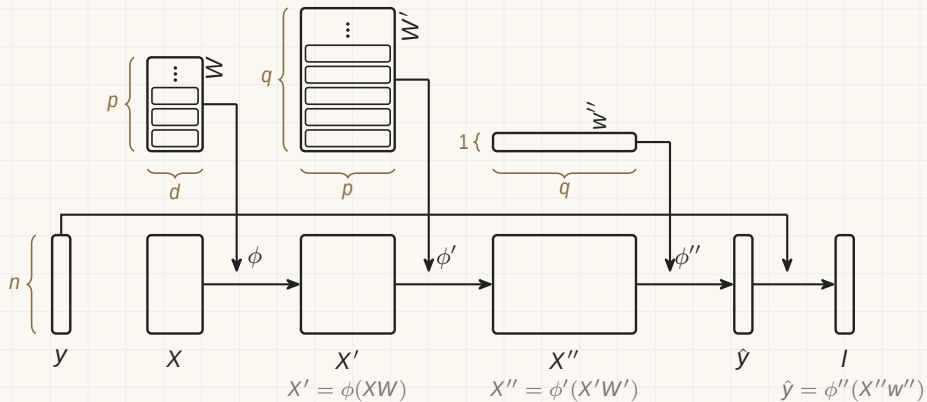
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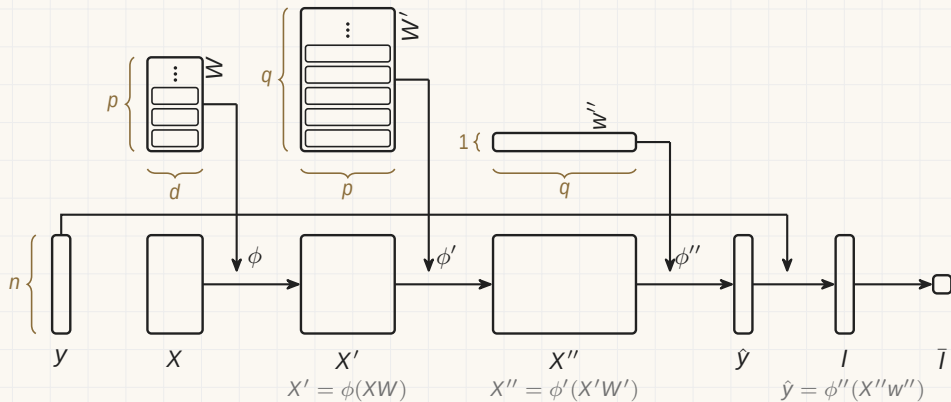
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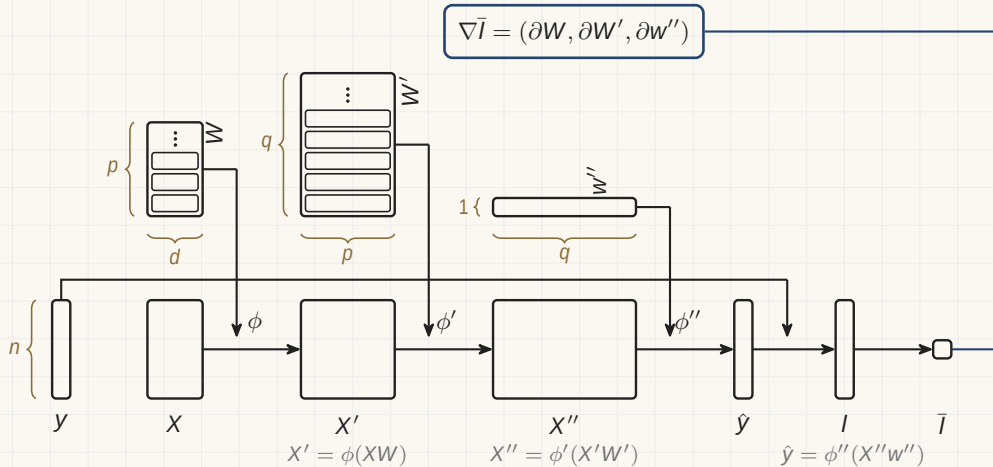
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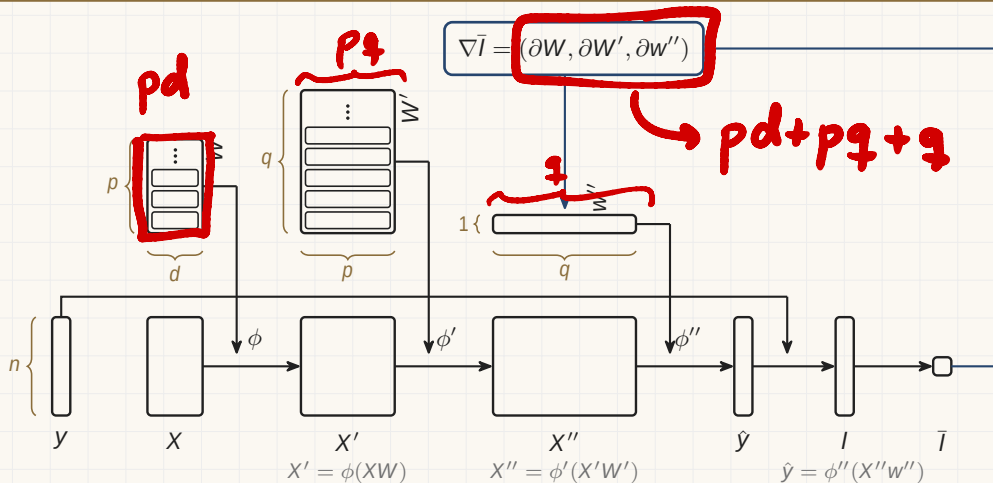


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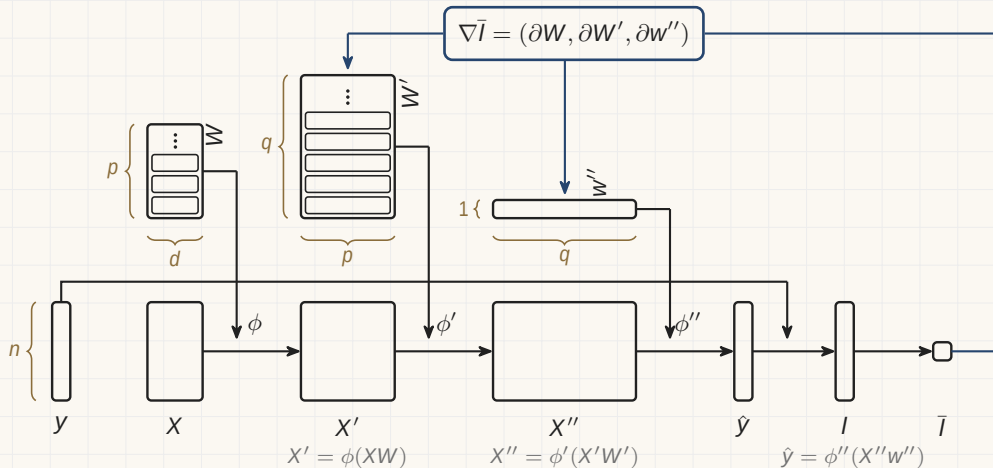


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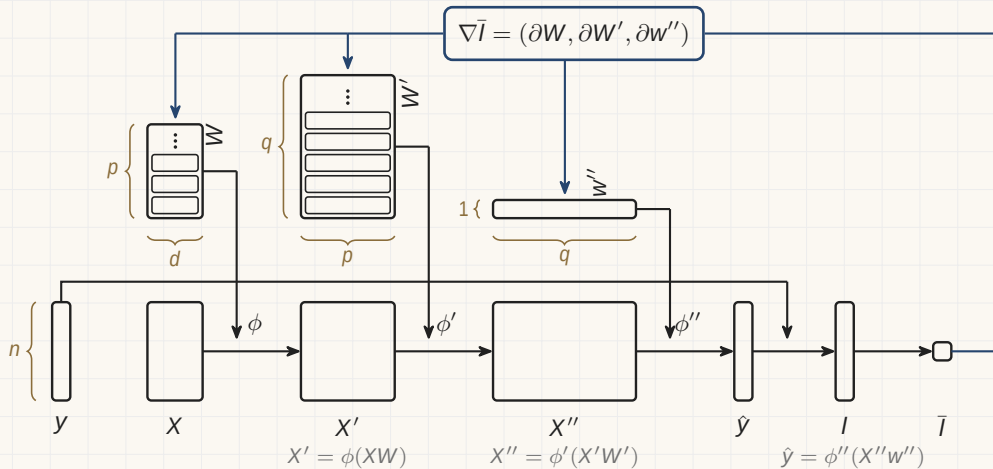
How many updates?



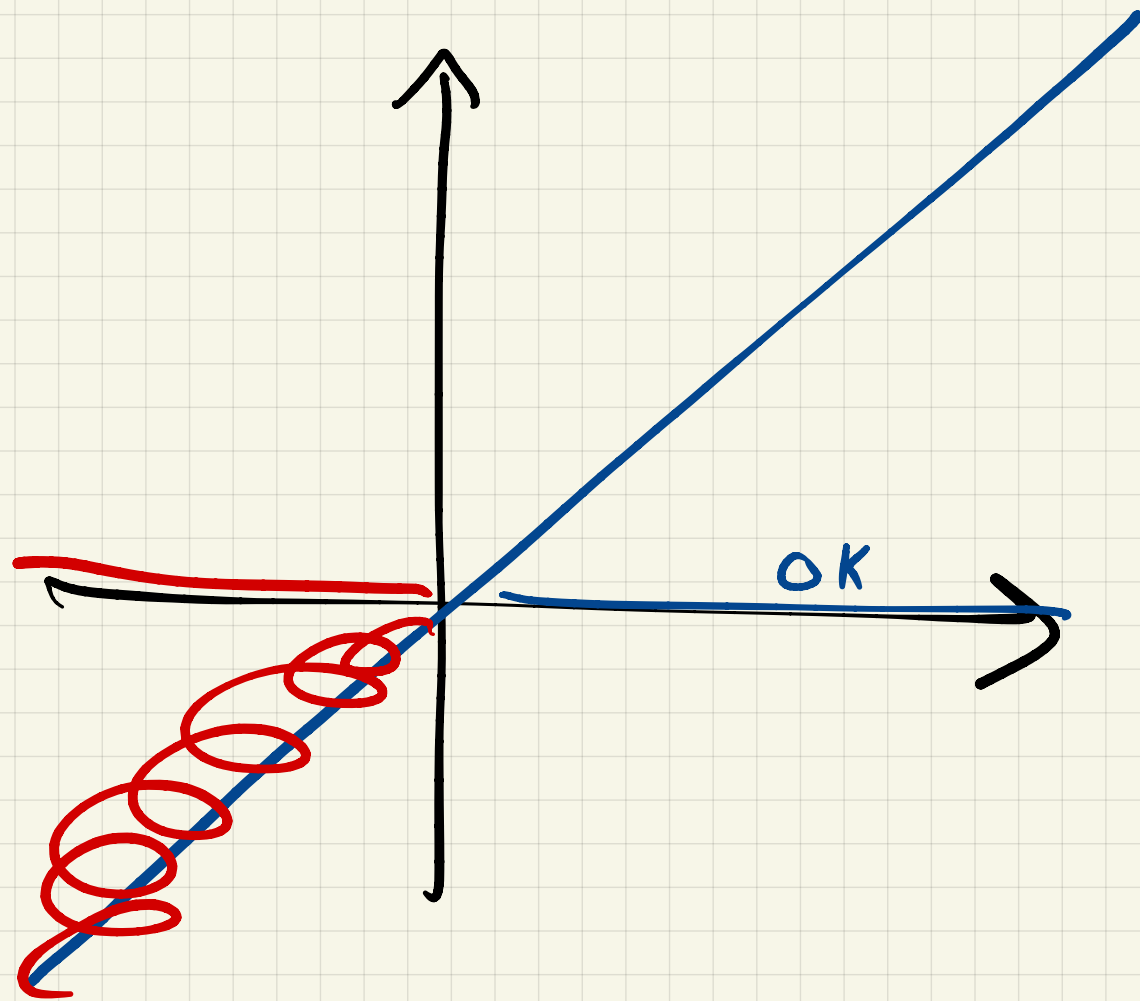
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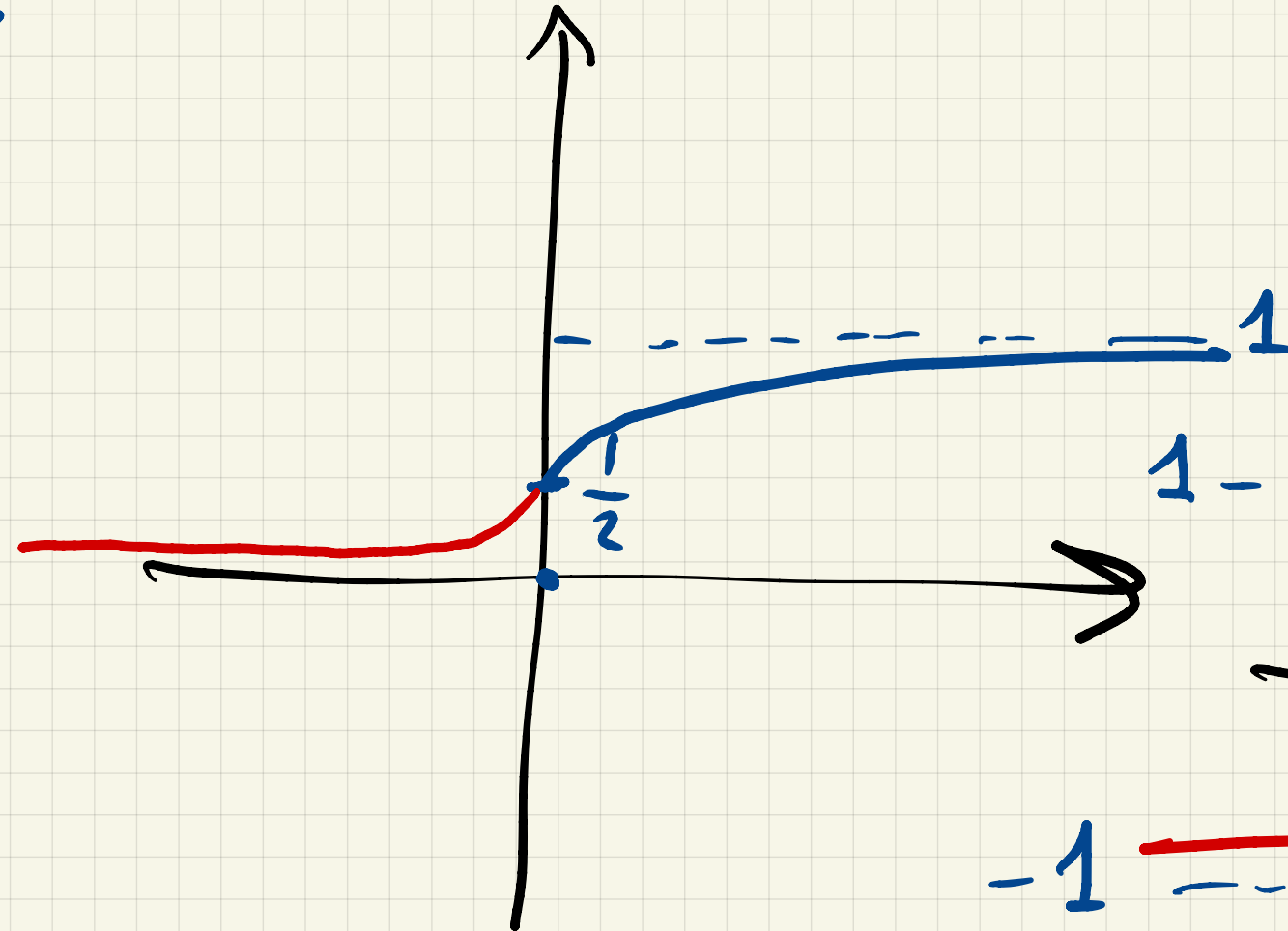
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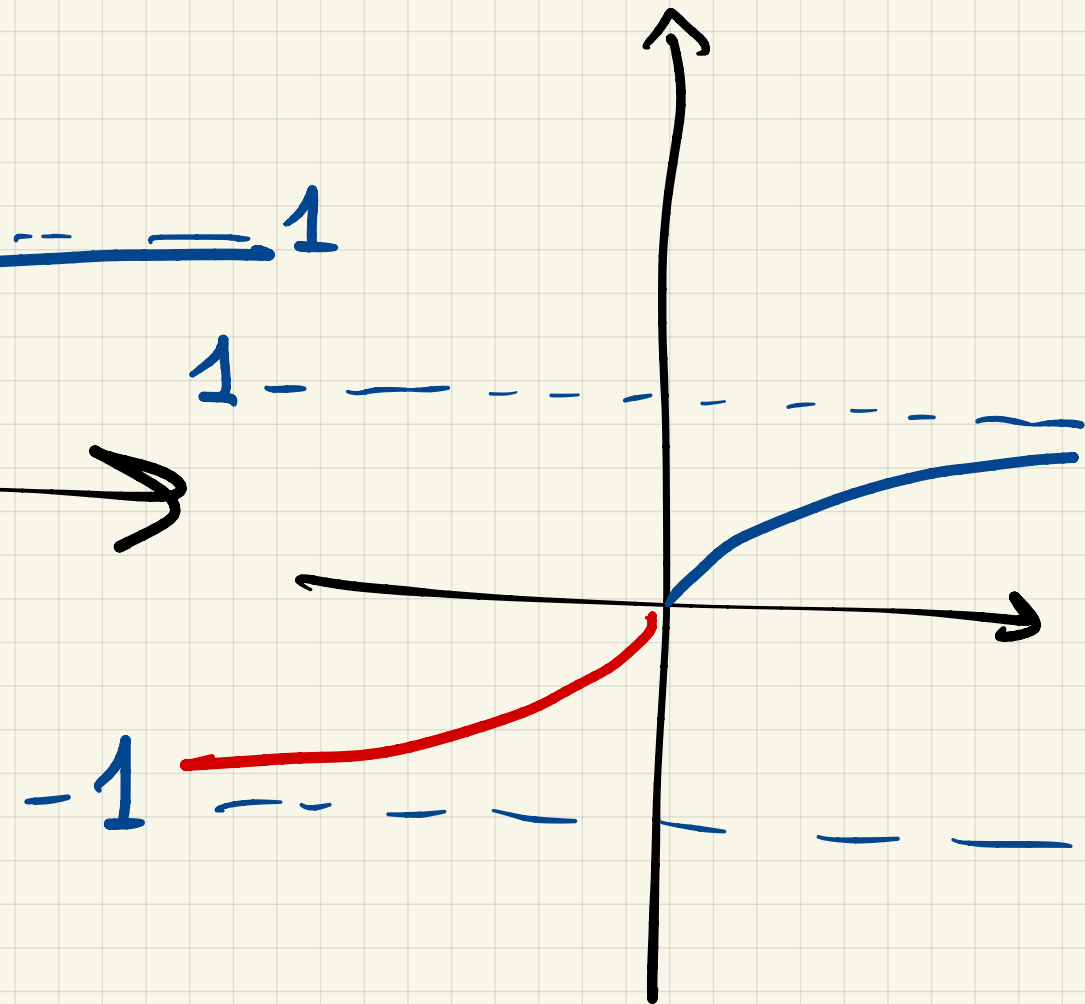
ReLU



Sigmoid



Tanh



The same network, drawn as neurons

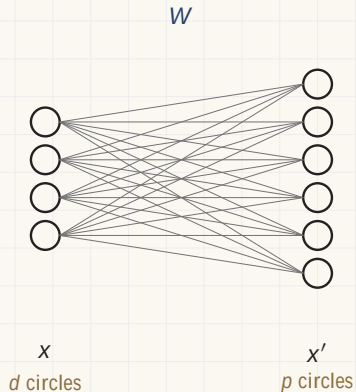
One circle for every column of every weight block.



x
 d circles

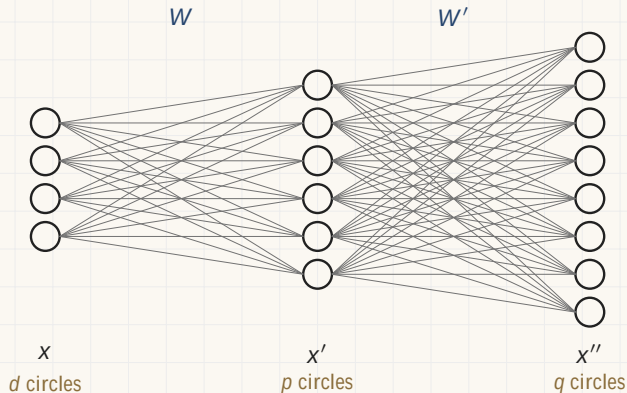
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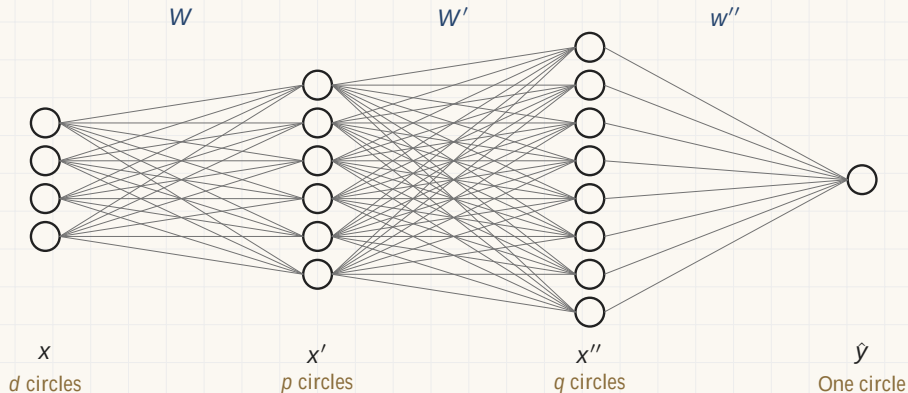
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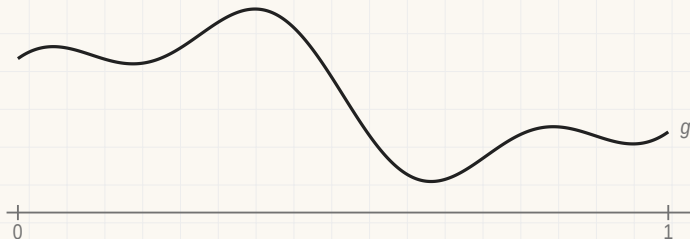
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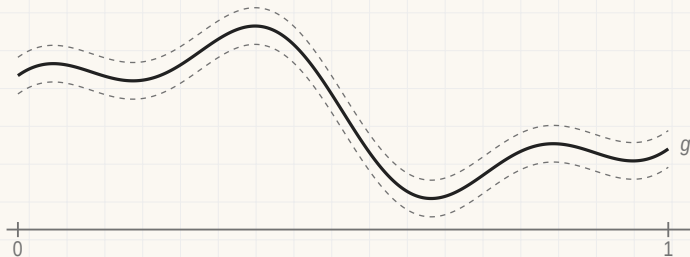
The universal approximation theorem

A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



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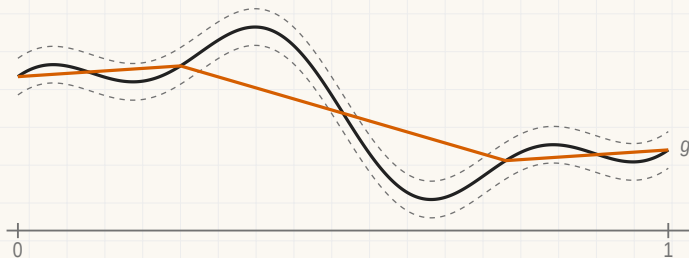
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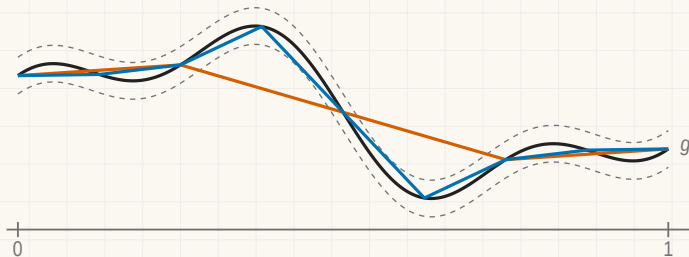
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$p = 4$, largest miss 0.85043

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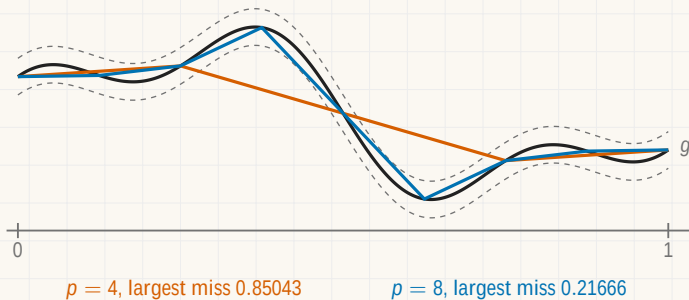
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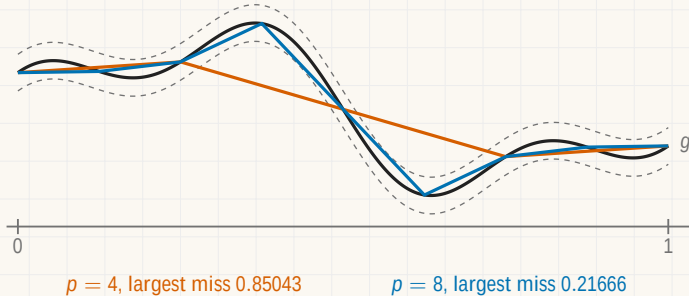
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A network with one hidden layer, and enough units in it, is a universal approximator.

The largest miss falls to 0.06944 at $p = 16$, to 0.01788 at $p = 32$ and to 0.00448 at $p = 64$.

Cybenko, 1989. Hornik, Stinchcombe and White, 1989.